Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

Contents

1 Population Principal Components **2**

2 Sample Principal Components and the SVD **3**

2.1 Sample PCs 3

2.2 Relation to the singular value decomposition 3

3 PCA and the Best Approximating Affine Subspace **3**

3.1 The problem 3

3.2 Observations 4

3.3 Correlation PCA 5

4 How Many Components to Keep? **5**

5 Example: Scores in Two Subjects Revisited **6**

Ch. 1, §6, introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition 7.1: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0\ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem 7.2: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers. **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u}, \qquad \frac{1}{2} \frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} - \sum_{j=1}^{k-1} \gamma_j \mathbf{u}_j = \mathbf{0}, \qquad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i < k)$: $\mathbf{u}_i^\top \boldsymbol{\Sigma} \mathbf{u} = \gamma_i$, and $\mathbf{u}_i^\top \boldsymbol{\Sigma} \mathbf{u} = \lambda_i \mathbf{u}_i^\top \mathbf{u} = 0$, so $\gamma_i = 0$. Hence $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k, \lambda_k$. Finally $\mathrm{Cov}(Z_j, Z_k) = \mathbf{u}_j^\top \boldsymbol{\Sigma} \mathbf{u}_k = \lambda_k \mathbf{u}_j^\top \mathbf{u}_k = 0$. ■

Theorem 7.3: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j) = \mathrm{tr}\boldsymbol{\Sigma} = \sum_{k=1}^p \lambda_k = \sum_k \mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k / \sum_j \lambda_j$.

2. $\mathrm{Corr}(Z_k, X_j) = \dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \mathrm{tr}\boldsymbol{\Lambda}$. (2) $\mathrm{Cov}(Z_k, X_j) = \mathbf{u}_k^\top \boldsymbol{\Sigma} \mathbf{e}_j = \lambda_k \mathbf{u}_k^\top \mathbf{e}_j = \lambda_k u_{jk}$, so $\mathrm{Corr} = \lambda_k u_{jk} / (\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem 7.4: Ky Fan maximum principle — q-dimensional subspaces

$$\max_{\mathbf{V} \in \mathbb{R}^{p \times q},\ \mathbf{V}^\top \mathbf{V} = \mathbf{I}_q} \mathrm{tr}(\mathbf{V}^\top \boldsymbol{\Sigma} \mathbf{V}) = \lambda_1 + \cdots + \lambda_q,$$

attained at $\mathbf{V} = [\mathbf{u}_1, \dots, \mathbf{u}_q]$ (and its rotations).

Proof. $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\operatorname{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{P}) \le \sum_i \lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P}) = \lambda_1 + \cdots + \lambda_q$. ■

Remark 7.5: PCA as optimal approximation

$\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu}) - \mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2 = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimize the expected squared reconstruction error, the minimum being $\lambda_{q+1} + \cdots + \lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}} = \mathbf{H}_n\mathbf{X} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centered data matrix, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\boldsymbol{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\ldots,k-1} \mathbf{u}^\top\mathbf{S}\mathbf{u}, \qquad \text{scores } t_{ik} = \hat{\mathbf{u}}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

2.2 Relation to the singular value decomposition

Write the SVD of the $p \times n$ matrix $\tilde{\mathbf{X}}^\top$ ($n > p$) as

$$\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \underbrace{[\mathbf{u}_1 \cdots \mathbf{u}_p]}_{p\times p} \underbrace{\begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_p \end{pmatrix}}_{p\times p} \underbrace{\begin{pmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_p^\top \end{pmatrix}}_{p\times n}, \qquad \mathbf{U}^\top\mathbf{U} = \mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V} = \mathbf{I}_p,\ d_1 \ge \cdots \ge d_p \ge 0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\operatorname{diag}\Big(\frac{d_1^2}{n-1}, \ldots, \frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix} \mathbf{z}_{(1)}^\top \\ \vdots \\ \mathbf{z}_{(p)}^\top \end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top \end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Theorem 7.6: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

3.1 The problem

Let the data be $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1, \ldots, p\}$. We wish to approximate $\mathbf{x}_1, \ldots, \mathbf{x}_n$ by $\bar{\mathbf{x}} + \mathbf{a}_1, \ldots, \bar{\mathbf{x}} + \mathbf{a}_n$ where $\operatorname{rank}([\mathbf{a}_1, \ldots, \mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}} + \mathbf{a}_i$ is better when the

error $\|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ by $\{\bar{\mathbf{x}} + \mathbf{a}_i\}$ is better when $\mathrm{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ $(p \times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r} \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \mathrm{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}} + \mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1 - \bar{\mathbf{x}}, \dots, \mathbf{x}_n - \bar{\mathbf{x}}]$ $(p \times n)$,

$$\sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$

because for a $p \times n$ matrix $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p \sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n \|\mathbf{b}_j\|_2^2 = \mathrm{tr}(\mathbf{B}\mathbf{B}^\top) = \mathrm{tr}(\mathbf{B}^\top\mathbf{B})$.

2. The sample PCA of $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ is given by

   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_r > \hat{\lambda}_{r+1} \geq \cdots \geq \hat{\lambda}_p$;

   (2) the SVD $\tilde{\mathbf{X}}_{(p\times n)}^\top = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i \hat{\mathbf{u}}_i \hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;

   (3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.

$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem 7.7: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimizes $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^r \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^p \hat{\lambda}_i$.

Proof. Step 1 (parametrize rank-r matrices). Since $\mathrm{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^n \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimizer is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top\mathbf{E})^{-1}\mathbf{E}^\top\tilde{\mathbf{x}}_i = \mathbf{E}^\top\tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto span($\mathbf{E}$) (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top \mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top \mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top \tilde{\mathbf{X}}^\top$.

Step 3 (profile out C). Plugging $\hat{\mathbf{C}}$ into ($\star\star$) (profiling, Theorem 2.22), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned} \min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top \tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\ &= \min_{\mathbf{E}} \left\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} \mathbf{E}\Big) \right\} \\ &= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1) \max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top). \end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Theorem 2.12) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimizer). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\operatorname{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r} \hat{\lambda}_i = \sum_{i>r} \hat{d}_i^2$. ■

Remark 7.8: interpretation

- The best approximating hyperplane is $\bar{\mathbf{x}} + \text{span}\{\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}} + \sum_{k \le r} (\hat{\mathbf{u}}_k^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i = \mathbf{E}^\top \tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x} = \bar{\mathbf{x}} + \sum s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimizing squared error and maximizing retained variance (the $\arg\max \mathbf{u}^\top \mathbf{S}\mathbf{u}$ definition) are the same problem, since $\text{SSE} = (n-1)\{\operatorname{tr}\mathbf{S} - \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r = \hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r > \hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardized data: $\mathbf{R} = \mathbf{D}_{\mathbf{\Sigma}}^{-1/2} \mathbf{\Sigma} \mathbf{D}_{\mathbf{\Sigma}}^{-1/2} = \mathbf{U}_R \mathbf{\Lambda}_R \mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\operatorname{tr}\mathbf{R} = p$, the proportion explained is λ_k^R / p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).

- **Kaiser's rule**: keep components with $\hat{\lambda}_k > \bar{\lambda} = \frac{1}{p}\sum_j \hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k > 1$.
- **Likelihood ratio testing**: under normality test $H_0(k) : \lambda_{k+1} = \cdots = \lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda = n(p-k)\log\frac{\bar{a}}{\bar{g}} \;\dot{\sim}\; \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a} \geq \bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

Remark 7.9: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \Rightarrow N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.